

Retrieve the Right Signal: Decomposition-Guided Retrieval-Augmented Forecasting for Financial Markets

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Abstract

Financial time-series forecasting is important for portfolio construction, risk management, and investment strategy design, but financial markets are highly nonstationary and often lack stable periodicity. Retrieval-augmented forecasting is attractive in this setting because similar market conditions can lead to similar historical market responses. However, when retrieval is performed over the raw signal, similarity can be dominated by level or trend rather than by detrended temporal patterns that are more relevant to future market reactions. We propose RISE (Retrieve the Right Signal: Decomposition-Guided Retrieval-Augmented Forecasting for Financial Markets), a decomposition-guided retrieval framework for financial time-series forecasting. RISE decomposes each input into a trend component and a detrended component using a strictly causal moving average, and performs retrieval in the detrended space. The retrieved detrended futures are used both as a direct retrieval-based forecasting component and as **Retrieval-as-Context (RAC)** tokens that provide additional context to the forecasting backbone. This design separates trend modeling from retrieval over detrended temporal patterns, reducing misleading raw-signal matches while preserving historical market responses that are relevant to the current query. It also allows retrieved information to influence representation learning, rather than being used only through output-level fusion. Across eight real-world financial

datasets and two TFB financial-market benchmarks, RISE outperforms retrieval-augmented, decomposition-based, attention-based, patch-based, clustering-based, and classical statistical baselines. On the eight real-world datasets, RISE ranks first in 23 out of 48 settings under both MSE and MAE; on the TFB benchmarks, it ranks first in 7 out of 8 MSE configurations.

CCS Concepts

• **Mathematics of computing** → **Time series analysis**; • **Computing methodologies** → *Neural networks*; • **Information systems** → *Information retrieval*.

Keywords

time series forecasting, retrieval-augmented forecasting, trend-seasonality decomposition, financial time series

ACM Reference Format:

Yongdam Kim, Hyeonji Park, Dongwoo Lee, Jeongseon Cho, Yein Lee, Jihoon Yang, and Sungwon Jung. 2026. Retrieve the Right Signal: Decomposition-Guided Retrieval-Augmented Forecasting for Financial Markets. In *Proceedings of the 35th ACM International Conference on Information and Knowledge Management (CIKM '26)*, November 07–11, 2026, Rome, Italy. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3799682.3841004>

1 Introduction

Recent advances in deep learning have substantially refined time-series forecasting. Traditional statistical models typically rely on explicit assumptions or structures to model autocorrelation, trend, volatility, and other temporal dependencies. Modern deep-learning models learn complex temporal dependencies through architectures such as MLPs, CNNs, Transformers, decomposition-based



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ACM ISBN 979-8-4007-2539-5/2026/11

<https://doi.org/10.1145/3799682.3841004>

models, and patch-based models. Three directions stand out in recent work. The first is decomposition-based forecasting (e.g., DLinear [19], Autoformer [18]), which separates a series into trend and seasonal or residual components and models the resulting components separately. The second is patch-based forecasting (e.g., PatchTST [12]), which slices the series into short segments and efficiently learns temporal interactions among nearby time steps. The third is retrieval-augmented forecasting (e.g., RAFT [7]), which retrieves past segments resembling the present and uses what followed them to inform the prediction.

Yet these approaches have largely been developed and validated on standard benchmarks with relatively stable periodicity, such as electricity, weather, and traffic. Financial series, by contrast, behave differently.

In financial forecasting [15, 16], predictive information is often less tied to the trend component or stable periodicity than to detrended temporal patterns around the trend [4, 6, 10]. Downstream tasks, including portfolio construction, volatility estimation [1, 5], risk management [4, 15, 16], and investment strategy design [15, 16], rely more heavily on detrended temporal patterns that reflect short-term market responses than on long-term price levels. Although such detrended temporal patterns rarely form stable periodic cycles, comparable historical conditions can lead to similar historical market responses.

This recurrence makes retrieval-based forecasting attractive for financial time series. By directly referring to past windows with similar detrended temporal patterns, retrieval mechanisms can use historical market responses more explicitly than models that store temporal structure only implicitly in their parameters. However, retrieval over the raw signal has a structural limitation in finance. Because financial trends vary substantially across macroeconomic regimes, raw-signal similarity can be dominated by level or trend rather than by detrended temporal patterns. Consequently, the retrieved windows may appear similar in overall level or trend while the retrieved detrended futures fail to align with the historical market response relevant to the current query. Figure 1 shows this failure mode on the S&P 500: raw-signal retrieval finds a higher-correlation neighbor whose future direction contradicts the ground truth. Detrended retrieval finds a lower-correlation neighbor whose future direction is aligned with the query.

To address these issues, we propose **RISE: Retrieve the Right Signal**, a decomposition-guided retrieval framework for financial time-series forecasting. The key idea is that retrieval should focus on detrended temporal patterns rather than on raw-signal similarity dominated by level or trend. RISE therefore decomposes the input into a trend component and a detrended component, and confines retrieval to the detrended space. This focuses retrieval on the detrended component, reducing misleading matches caused by raw-signal similarity dominated by level or trend while preserving detrended temporal patterns.

RISE further performs multi-scale retrieval to capture detrended temporal patterns at different temporal resolutions. The retrieved detrended futures are aggregated by similarity and used in two complementary ways: as a retrieval-based forecasting component added directly to the final prediction, and as **Retrieval-as-Context (RAC)** tokens that serve as additional context for the forecasting

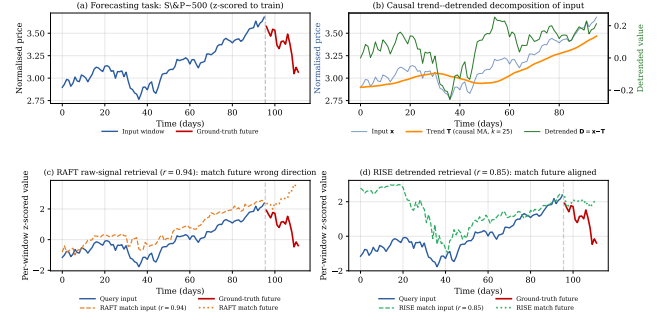


Figure 1: Motivating example on the S&P 500 index (16-day forecast horizon). Panels (c) and (d) plot the query alongside each pathway’s nearest neighbor (normalized via per-window z-scoring such that visual shape similarity aligns with the Pearson correlation r). The complete RISE and RAFT models, which fuse retrieval with trend and patch predictions, are evaluated in Section 4. RAFT’s raw-signal retrieval (c) yields a high-correlation neighbor whose future trajectory contradicts the ground truth (GT ↓, match ↑). In contrast, RISE’s detrended retrieval (d) identifies a neighbor with a slightly lower correlation whose future aligns accurately with the ground truth (both ↓). The example suggests that retrieval similarity in financial time series should emphasize detrended temporal patterns rather than raw-signal similarity dominated by level or trend.

backbone’s representation learning. Through this decomposition-guided design, RISE jointly models stable trend information and detrended temporal patterns, enabling more robust forecasting even under the highly nonstationary conditions of financial markets.

Contributions.

- We propose decomposition-guided retrieval for financial time-series forecasting. RISE decomposes each series into a trend component and a detrended component, and performs retrieval in the detrended space to reduce misleading matches caused by raw-signal similarity dominated by level or trend.
- RISE incorporates multi-scale retrieval over the detrended component, allowing the model to compare detrended temporal patterns at different temporal resolutions and aggregate the retrieved detrended futures by similarity.
- We introduce Retrieval-as-Context (RAC), which projects retrieved detrended futures into context tokens and feeds them into the forecasting backbone.
- RISE achieves state-of-the-art performance across diverse financial benchmarks, including eight real-world datasets and two TFB financial-market benchmarks (NASDAQ and NYSE).

2 Related Work

Financial time-series forecasting has long been studied through both traditional statistical models and deep-learning approaches. ARIMA models capture autocorrelation and trend-based patterns, ARCH/GARCH models describe conditional volatility [1, 5], and

VAR models capture dependencies among macroeconomic variables [15]. Random-walk and random-walk-with-drift baselines have also served as simple but strong references, reflecting the efficient-market view that market prices already incorporate available information [6, 10] and illustrating the empirical difficulty of predicting financial markets [11]. Deep-learning models such as RNNs and LSTMs have been used to learn temporal dependencies in financial time series, while recent approaches such as MASTER [8], MATCC [3], and DoubleAdapt [20] have advanced cross-sectional stock return prediction and ranking-oriented settings.

Recent deep time-series forecasting models have explored several architectural directions. Decomposition-based methods such as DLinear [19] and Autoformer [18] separate a series into trend and seasonal or residual components and model the resulting components separately. These methods make component separation explicit, but they mainly use the separated components for forecasting rather than for deciding which component should serve as the basis for historical comparison. Patch-based models such as PatchTST [12] divide the input into short segments to learn temporal interactions among nearby time steps. Transformer-based and multiscale architectures, including iTransformer [9] and TimeMixer [17], further improve forecasting by redesigning attention or mixing mechanisms for long-range temporal modeling. DUET [14] is also related as a clustering-based multivariate forecasting method that exploits structured grouping in time-series data. These architectures improve representation or structural modeling, but they do not explicitly retrieve similar historical market responses in financial time series.

Retrieval-augmented forecasting provides another direction by connecting the current segment with similar past segments and using what followed them to inform the prediction. RAFT [7] is a representative method in this line, retrieving past windows resembling the present and incorporating the corresponding future information into the forecast. GTR [2] is also related in that it exploits historical patterns or memory, although it is closer to using learned global cycle representations than to explicit nearest-neighbor retrieval. These methods introduce historical information into forecasting, but they do not explicitly address which signal component should be used for retrieval in financial time series, where raw-signal similarity can be dominated by level or trend.

Unlike these prior approaches, RISE combines decomposition and retrieval by using the detrended component as the retrieval target in financial time-series forecasting. This design distinguishes RISE from decomposition-based models that use separated components mainly for forecasting, patch-based models that encode temporal patterns implicitly, and retrieval-based models that typically retrieve from the raw signal or raw-signal-derived representations.

3 Proposed Method

RISE (Figure 2) uses decomposition as a routing mechanism. A strictly causal moving-average filter splits the input X into a slow trend T and a detrended component $D = X - T$: T is routed to a channel-independent linear head, while D is processed in parallel by a nearest-neighbor retrieval module and a patch-based Transformer. This confines retrieval to the component where reusable patterns

reside, avoiding level-dominated similarity. The two detrended modules are coupled through **Retrieval-as-Context (RAC)**: retrieved detrended futures are injected as context tokens into the patch encoder, so that retrieval shapes representation learning rather than only output-level fusion. A learnable weighted fusion of the three pathway outputs produces the final forecast.

3.1 Problem Formulation and Notation

Given a multivariate look-back window $\mathbf{x}^{\text{raw}} \in \mathbb{R}^{L \times C}$, RISE produces forecasts for all C channels over horizon H . We denote the all-channel future window by $\mathbf{y}_{\text{all}}^{\text{raw}} \in \mathbb{R}^{H \times C}$.

Let $C_y \subseteq \{1, \dots, C\}$ denote the set of channels used for supervision and evaluation, and let $C' = |C_y|$. The supervised target is

$$\mathbf{y}^{\text{raw}} = \mathbf{y}_{\text{all}, C_y}^{\text{raw}} \in \mathbb{R}^{H \times C'}. \quad (1)$$

In the fully multivariate setting, $C_y = \{1, \dots, C\}$ and thus $C' = C$. In the multivariate-to-single (MS) setting, $C_y = \{c^*\}$ and $C' = 1$, where c^* denotes the target channel. In our financial datasets, the target variable is the closing price, and the target column is placed as the last channel; hence $c^* = C$.

We distinguish three versions of the same window. Raw observations are denoted by $\mathbf{x}^{\text{raw}}$ and $\mathbf{y}^{\text{raw}}$. Globally standardized observations are denoted by $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{y}}$. Locally anchored inputs and targets, which are the actual quantities processed by RISE, are denoted by $\mathbf{x}$ and $\mathbf{y}$ without tildes. The index t denotes time, c denotes channel, and $\tilde{\mathbf{x}}_{t,c}$ is the last observed standardized input value in channel c .

3.2 Two-Stage Normalization

Financial time series exhibit both long-term scale drift and local level shifts. We address these effects separately. First, each input channel is standardized using statistics computed from the training split:

$$\tilde{\mathbf{x}}_{t,c} = \frac{\mathbf{x}_{t,c}^{\text{raw}} - \mu_c^{\text{train}}}{\sigma_c^{\text{train}}}, \quad c = 1, \dots, C. \quad (2)$$

The all-channel future window is standardized with the same training statistics:

$$\tilde{\mathbf{y}}_{\text{all},t,c} = \frac{\mathbf{y}_{\text{all},t,c}^{\text{raw}} - \mu_c^{\text{train}}}{\sigma_c^{\text{train}}}, \quad c = 1, \dots, C. \quad (3)$$

The standardized supervised target is the subset $\tilde{\mathbf{y}}_{t,c} = \tilde{\mathbf{y}}_{\text{all},t,c}$ for $c \in C_y$. In the MS setting, this selects only the closing-price channel $c^* = C$.

Second, for each forecasting window, we subtract the last observed standardized input value as a local anchor. For the input window, this operation is applied to all input channels:

$$\mathbf{x}_{t,c} = \tilde{\mathbf{x}}_{t,c} - \tilde{\mathbf{x}}_{L,c}, \quad c = 1, \dots, C. \quad (4)$$

For the future target, the same operation is applied only to the target channels:

$$\mathbf{y}_{t,c} = \tilde{\mathbf{y}}_{t,c} - \tilde{\mathbf{x}}_{L,c}, \quad c \in C_y. \quad (5)$$

In the MS setting, Eq. (5) becomes

$$\mathbf{y}_{t,c^*} = \tilde{\mathbf{y}}_{t,c^*} - \tilde{\mathbf{x}}_{L,c^*}, \quad c^* = C. \quad (6)$$

Thus, the model learns the future deviation of the target variable relative to the last observed value of the same target channel.

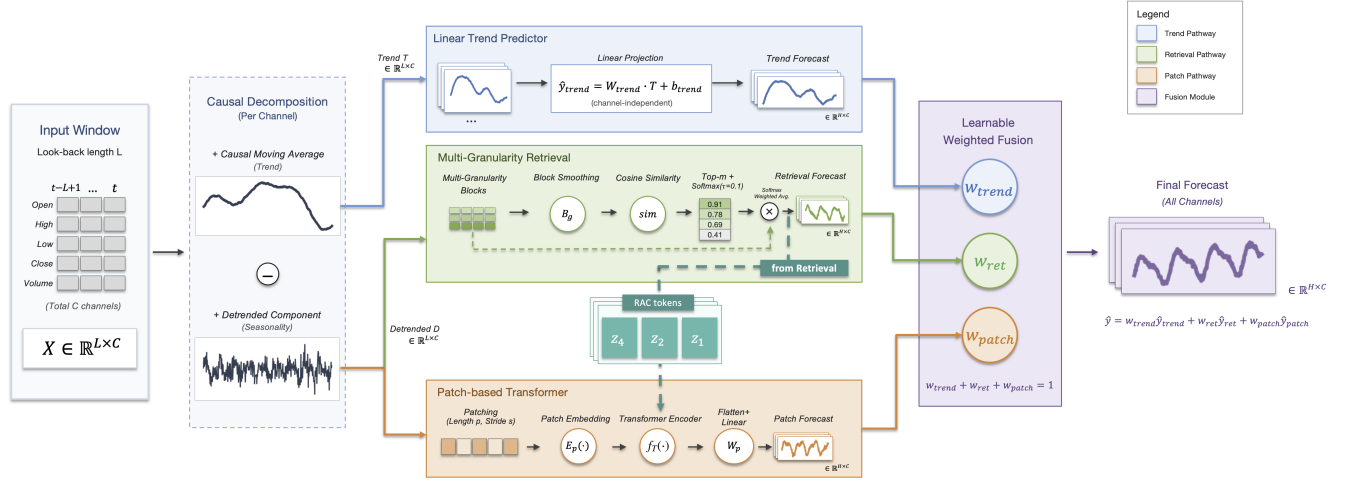


Figure 2: RISE architecture. A strictly causal moving average splits the input X into a macro trend T and a detrended signal $D = X - T$. Three pathways run in parallel—trend (blue), retrieval (green) and patch (orange)—and their forecasts are combined by a learnable weighted fusion (purple). Retrieval-as-Context (RAC) tokens carry the retrieved context into the patch encoder (Figure 3). Hyperparameters and the per-window offset are described in Section 3.

3.3 Causal Decomposition and Trend Pathway

RISE decomposes the locally anchored input $x \in \mathbb{R}^{L \times C}$ into a causal trend component T and a detrended component D . The trend is extracted using a strictly causal moving average. To compute the trend without using future observations, RISE averages only the current and previous values. For early time steps, where fewer than k past observations are available, we extend the sequence to the left by repeating the first value $x_{1,c}$. Equivalently, define

$$x_{t,c}^{\text{pad}} = \begin{cases} x_{1,c}, & t < 1, \\ x_{t,c}, & 1 \leq t \leq L. \end{cases} \quad (7)$$

The causal moving-average trend is then

$$T_{t,c} = \frac{1}{k} \sum_{i=0}^{k-1} x_{t-i,c}^{\text{pad}}. \quad (8)$$

This filter never accesses $x_{t+i,c}$ for $i > 0$, so it cannot leak future information. The remaining high-frequency component is the detrended signal:

$$D = x - T. \quad (9)$$

The trend pathway predicts the future contribution of the locally anchored low-frequency trend using a linear projection:

$$\hat{y}_{:,c}^{(\text{trend})} = W_{\text{trend}} T_{:,c} + b_{\text{trend}}. \quad (10)$$

Here, $T_{:,c} \in \mathbb{R}^L$ denotes the observed trend vector for channel c , $W_{\text{trend}} \in \mathbb{R}^{H \times L}$ and $b_{\text{trend}} \in \mathbb{R}^H$ are the channel-shared projection parameters, and $\hat{y}_{:,c}^{(\text{trend})} \in \mathbb{R}^H$ denotes the trend-pathway forecast over the prediction horizon.

3.4 Multi-Granularity Seasonal Retrieval

The retrieval pathway searches for historical patterns using only the detrended signal D . We define a set of retrieval granularities $\mathcal{G}$,

where each $g \in \mathcal{G}$ controls the temporal resolution of the search. In our default setting, $\mathcal{G} = \{4, 2, 1\}$. We choose L and H such that they are divisible by every $g \in \mathcal{G}$. This allows block-level smoothing to be applied without padding or truncation.

Before training, we construct a memory bank from the training split. For each training window j , we store a detrended input key and a detrended future value. Unlike the patch and trend pathways, the retrieval bank is constructed from globally standardized windows before local anchoring. Specifically, the standardized input window $\tilde{x}^{(j)} \in \mathbb{R}^{L \times C}$ and standardized future window $\tilde{y}_{\text{all}}^{(j)} \in \mathbb{R}^{H \times C}$ are detrended separately:

$$D^{(j)} = \tilde{x}^{(j)} - \text{MA}_k(\tilde{x}^{(j)}), \quad D_y^{(j)} = \tilde{y}_{\text{all}}^{(j)} - \text{MA}_k(\tilde{y}_{\text{all}}^{(j)}). \quad (11)$$

Here, $D^{(j)}$ is used as the retrieval key, and $D_y^{(j)}$ is used as the future residual associated with that key.

For any detrended input sequence $D \in \mathbb{R}^{L \times C}$, either from the online query or from the memory bank, we first form a block-level sequence $B_g \in \mathbb{R}^{(L/g) \times C}$ by averaging every block of g time steps:

$$B_{g,b,c} = \frac{1}{g} \sum_{r=1}^g D_{(b-1)g+r,c}, \quad b = 1, \dots, L/g. \quad (12)$$

We then repeat each block average g times along the temporal axis to obtain the smoothed sequence $D_g \in \mathbb{R}^{L \times C}$, and subtract its terminal value:

$$D_{g,t,c} \leftarrow D_{g,t,c} - D_{g,L,c}. \quad (13)$$

The same block-smoothing and terminal-recentering operations are applied to the memory input $D^{(j)}$ to obtain $D_g^{(j)}$.

For the future residual stored in the memory bank, we apply block smoothing at the same granularity g . Specifically, for $D_y^{(j)} \in$

$\mathbb{R}^{H \times C}$, we compute

$$\mathbf{B}_{g,y,b,c}^{(j)} = \frac{1}{g} \sum_{r=1}^g \mathbf{D}_{y,(b-1)g+r,c}^{(j)}, \quad b = 1, \dots, H/g. \quad (14)$$

Repeating each future block average g times along the temporal axis gives a smoothed future residual. As with the retrieval keys, we subtract its terminal value:

$$\mathbf{D}_{g,y,t,c}^{(j)} \leftarrow \mathbf{D}_{g,y,t,c}^{(j)} - \mathbf{D}_{g,y,H,c}^{(j)}. \quad (15)$$

The resulting $\mathbf{D}_{g,y}^{(j)} \in \mathbb{R}^{H \times C}$ is constant within each block of size g and serves as the retrieved future residual at granularity g .

For similarity search, we flatten the smoothed query and memory-bank entries:

$$\mathbf{q}_g = \text{vec}(\mathbf{D}_g), \quad \mathbf{k}_g^{(j)} = \text{vec}(\mathbf{D}_g^{(j)}). \quad (16)$$

The similarity score is mean-subtracted cosine similarity:

$$\text{sim}_g(j) = \left\langle \frac{\mathbf{q}_g - \bar{\mathbf{q}}_g}{\max(\|\mathbf{q}_g - \bar{\mathbf{q}}_g\|, \epsilon)}, \frac{\mathbf{k}_g^{(j)} - \bar{\mathbf{k}}_g^{(j)}}{\max(\|\mathbf{k}_g^{(j)} - \bar{\mathbf{k}}_g^{(j)}\|, \epsilon)} \right\rangle. \quad (17)$$

Here, $\bar{\mathbf{q}}_g$ and $\bar{\mathbf{k}}_g^{(j)}$ denote the scalar means of the flattened vectors, and ϵ is a small constant for numerical stability. During training, memory entries whose input-future intervals overlap with the query interval are masked out. Equivalently, if two training windows start at indices i and j , the memory entry is excluded whenever the intervals $[i, i + L + H - 1]$ and $[j, j + L + H - 1]$ overlap.

Let $\mathcal{N}_g$ be the top- m retrieved neighbors under sim_g . Their weights are computed by a temperature-scaled softmax:

$$a_g(j) = \frac{\exp(\text{sim}_g(j)/\tau)}{\sum_{j' \in \mathcal{N}_g} \exp(\text{sim}_g(j')/\tau)}, \quad j \in \mathcal{N}_g. \quad (18)$$

The retrieved future at granularity g is the weighted average of the corresponding block-smoothed future residuals:

$$\bar{\mathbf{y}}_g = \sum_{j \in \mathcal{N}_g} a_g(j) \mathbf{D}_{g,y}^{(j)}. \quad (19)$$

Thus, $\bar{\mathbf{y}}_g \in \mathbb{R}^{H \times C}$ denotes the retrieved detrended future at granularity g .

3.5 Patch Transformer with Retrieval-as-Context

The patch pathway models local nonlinear dynamics in the detrended signal $\mathbf{D}$. We split the sequence into overlapping patches of length p and stride s . Since the implementation appends replication padding of length s to the right side of the sequence before patching, the number of patches is

$$N_p = \left\lfloor \frac{L + s - p}{s} \right\rfloor + 1. \quad (20)$$

Each patch is projected into a d -dimensional token and augmented with sinusoidal positional encoding, producing patch tokens $\mathbf{u}_{1,c}, \dots, \mathbf{u}_{N_p,c}$ for channel c .

RISE injects retrieval information through Retrieval-as-Context (RAC), illustrated in Figure 3. For each granularity g , the retrieved future $\bar{\mathbf{y}}_g$ is projected into one context token per channel:

$$\mathbf{c}_{g,c} = \mathbf{W}_g^{\text{ctx}} \bar{\mathbf{y}}_{g,c} + \mathbf{b}_g^{\text{ctx}} + \mathbf{e}_g. \quad (21)$$

Here $\mathbf{W}_g^{\text{ctx}} \in \mathbb{R}^{d \times H}$ and $\mathbf{b}_g^{\text{ctx}}, \mathbf{e}_g \in \mathbb{R}^d$. The projection parameters are shared across channels and differ only by granularity g . In the implementation, RAC tokens are appended after the patch tokens:

$$\mathbf{S}_c = [\mathbf{u}_{1,c}, \dots, \mathbf{u}_{N_p,c}, \mathbf{c}_{g1,c}, \dots, \mathbf{c}_{g|G|,c}]. \quad (22)$$

The sequence $\mathbf{S}_c$ is processed by a two-layer Transformer encoder with full self-attention. After encoding, only the original patch-token positions are retained:

$$\mathbf{H}_c = \text{Encoder}(\mathbf{S}_c)_{1:N_p}. \quad (23)$$

The retained patch representations are flattened and projected to the forecast horizon:

$$\hat{\mathbf{y}}_{:,c}^{(\text{patch})} = \mathbf{W}_p \text{vec}(\mathbf{H}_c) + \mathbf{b}_p. \quad (24)$$

3.6 Weighted Fusion and Training Objective

The retrieval pathway also provides a direct retrieval forecast. Since $\bar{\mathbf{y}}_g$ is block-smoothed, it contains only H/g unique temporal values. We keep one value per block as $\mathbf{p}_g \in \mathbb{R}^{(H/g) \times C}$ and apply a granularity-specific linear head:

$$\hat{\mathbf{y}}_{g,::c}^{(\text{ret})} = \mathbf{W}_g \mathbf{p}_{g,::c} + \mathbf{b}_g, \quad \mathbf{W}_g \in \mathbb{R}^{H \times H/g}, \quad \mathbf{b}_g \in \mathbb{R}^H. \quad (25)$$

The retrieval-pathway output is the sum of the granularity-specific predictions:

$$\hat{\mathbf{y}}^{(\text{ret})} = \sum_{g \in \mathcal{G}} \hat{\mathbf{y}}_g^{(\text{ret})}. \quad (26)$$

The three pathway outputs are first computed for all channels and then combined by a learnable weighted fusion:

$$\hat{\mathbf{y}}_{\text{all},t,c} = w_{\text{patch}} \hat{\mathbf{y}}_{t,c}^{(\text{patch})} + w_{\text{trend}} \hat{\mathbf{y}}_{t,c}^{(\text{trend})} + w_{\text{ret}} \hat{\mathbf{y}}_{t,c}^{(\text{ret})}, \quad (27)$$

for $c = 1, \dots, C$. The locally anchored forecast is converted back to the globally standardized target space by adding back the corresponding local anchor:

$$\hat{\mathbf{y}}_{t,c} = \hat{\mathbf{y}}_{\text{all},t,c} + \tilde{\mathbf{x}}_{L,c}, \quad c \in \mathcal{C}_y. \quad (28)$$

RISE is trained using the forecasting MSE in the globally standardized target space. Although the model predicts locally anchored deviations, the prediction is first restored to the globally standardized level using Eq. (28). The loss is then computed only over the target channels:

$$\mathcal{L} = \frac{1}{H|\mathcal{C}_y|} \sum_{t=1}^H \sum_{c \in \mathcal{C}_y} (\hat{\mathbf{y}}_{t,c} - \tilde{\mathbf{y}}_{t,c})^2. \quad (29)$$

In the MS setting, $\mathcal{C}_y = \{c^*\}$ with $c^* = C$, and the loss reduces to

$$\mathcal{L}_{\text{MS}} = \frac{1}{H} \sum_{t=1}^H (\hat{\mathbf{y}}_{t,c^*} - \tilde{\mathbf{y}}_{t,c^*})^2. \quad (30)$$

Thus, in the financial MS setting, RISE may process all input channels internally, but supervision and evaluation are performed only on the closing price channel.

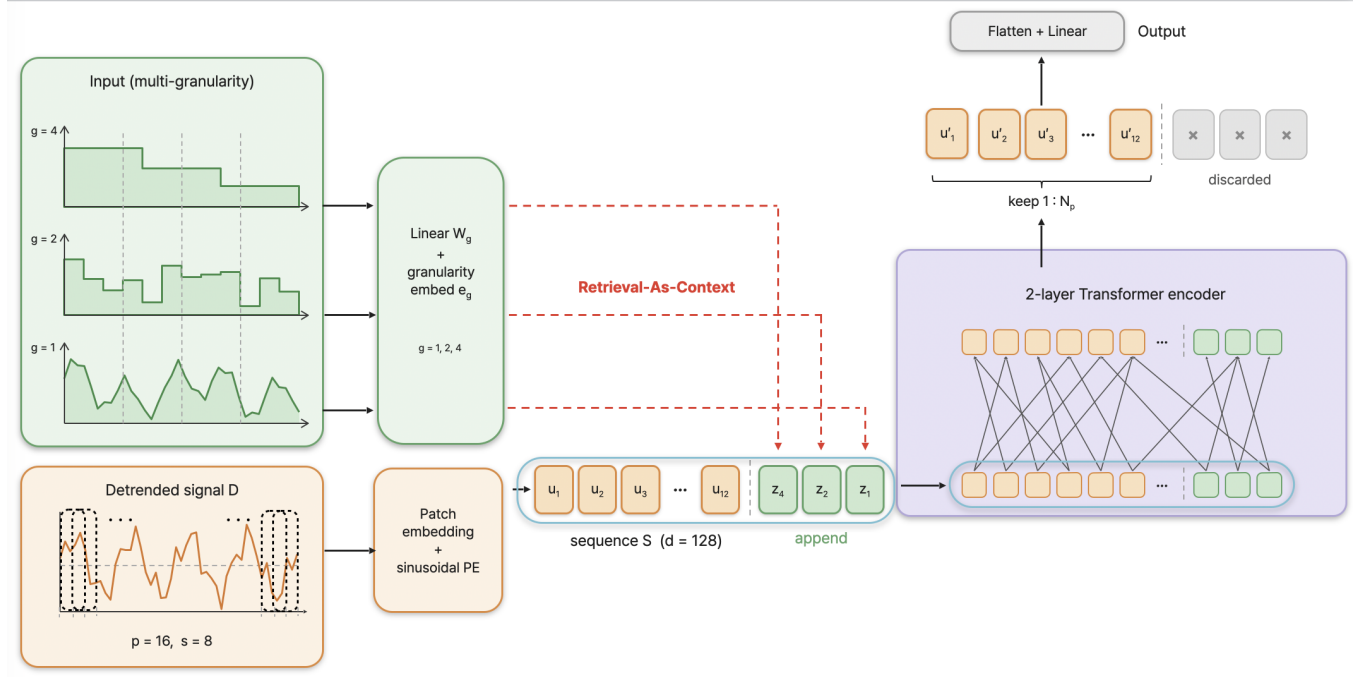


Figure 3: Retrieval-as-Context (RAC). The retrieved detrended future at each granularity is projected into one context token per channel (green, drawn as z_g) and appended after the patch tokens obtained from D (orange). Full self-attention in the 2-layer encoder exposes every patch token to the retrieved context; the context positions are then discarded and only the N_p patch positions are decoded.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate RISE on two complementary testbeds with distinct objectives: (1) a curated real-world financial forecasting suite designed to capture heterogeneous market dynamics, and (2) financial-domain datasets selected from the public Time-Series Forecasting Benchmark (TFB) [13] for standardized external validation (Table 1). Our real-world suite consists of eight financial datasets spanning four asset classes and four global regions (US, JP, HK, DE). We adopt the multivariate-to-univariate (‘MS’) forecasting setting, where the model predicts the future value of a target asset using OHLCV features (Open, High, Low, Close, Volume). The suite spans equity indices across multiple geographies (S&P 500, NASDAQ, Nikkei 225, Hang Seng, DAX), commodities (Gold), foreign exchange markets (EUR/USD) [11], and cryptocurrency markets (Bitcoin), covering a diverse range of market microstructures — from near-random-walk FX dynamics to highly non-stationary crypto volatility — allowing us to assess RISE across heterogeneous financial market conditions. For external validation, we further evaluate RISE on two financial-domain datasets from TFB [13]: NASDAQ and NYSE, following the official multivariate-to-multivariate (‘M’) setting.

Baselines. We benchmark RISE against seven state-of-the-art forecasting models covering the main architectural directions discussed above: retrieval-augmented forecasting, decomposition-based modeling, patch-based modeling, Transformer-based and

Table 1: Dataset characteristics. The eight real-world datasets span four asset classes and four geographies, cover daily observations from 2015 to 2025 collected from Yahoo Finance, and are evaluated under the multivariate-to-univariate (‘MS’) setting using OHLCV features. The two TFB datasets are financial-domain benchmarks evaluated under the official multivariate-to-multivariate (‘M’) setting. All datasets are sampled at a daily frequency and evaluated with chronological 7:1:2 train/validation/test splits.

Dataset	Vars	Samples	Split	Period	Domain
<i>Real-world financial datasets (ours)</i>					
S&P 500	5	2,517	7:1:2	2015–2025	US Equity
NASDAQ	5	2,517	7:1:2	2015–2025	US Equity
Nikkei 225	5	2,443	7:1:2	2015–2025	JP Equity
Hang Seng	5	2,513	7:1:2	2015–2025	HK Equity
DAX	5	2,593	7:1:2	2015–2025	DE Equity
Gold	5	2,517	7:1:2	2015–2025	Commodity
EUR/USD	5	2,609	7:1:2	2015–2025	Forex
Bitcoin	5	2,517	7:1:2	2015–2025	Crypto
<i>TFB benchmark datasets [13]</i>					
NASDAQ	5	1,240	7:1:2	—	Stock
NYSE	5	1,239	7:1:2	—	Stock

multiscale architectures, and clustering-based multivariate forecasting. Specifically, we compare against RAFT [7], GTR [2], DLinear [19], PatchTST [12], iTransformer [9], TimeMixer [17], and DUET [14]. We additionally include three classical statistical baselines, namely Persistence, Random Walk with Drift, and ARIMA(1,1,1), to establish random-walk reference bounds.

Evaluation protocol and data leakage mitigation. To ensure a fair comparison across all evaluated models, the two-stage normalization detailed in Section 3 is applied identically: global per-channel z-score statistics are computed strictly on the training partition, and the terminal input value is subtracted locally during sample construction.

For RISE specifically, three additional measures strictly preclude look-ahead bias. First, the offline memory bank is populated exclusively with sliding windows from the training split, ensuring that validation and test queries can only retrieve historically observed memory bank entries. Second, during training, memory entries whose input-future intervals overlap with the query window are excluded from retrieval, preventing any query from accessing temporally adjacent training windows (Section 3.4). Third, detrended retrieved futures are precomputed offline employing the identical deterministic causal moving average (Eq. 8) utilized during online inference, ensuring that neither the decomposition process nor the retrieval candidates are exposed to validation or test data.

Implementation details. All deep-learning experiments are conducted on two NVIDIA RTX 3090 GPUs, using the Adam optimizer with a learning rate of 10^{-4} , a batch size of 32, and a CosineAnnealingWarmRestarts scheduler over 10 epochs. We evaluate predictive horizons $H \in \{8, 16, 24, 36, 48, 60\}$, corresponding approximately to 1.5 weeks to 3 months.

To reduce evaluation asymmetries from disparate hyperparameter tuning budgets, Table 2 evaluates all deep-learning models under a controlled protocol with a fixed look-back length ($L=96$) and reports averages over five random seeds $\{0, 1, 2, 3, 4\}$. For RISE and RAFT, top- m is selected via grid search over $\{1, 5, 10, 20\}$ and fixed to 20 in the main results; for RAFT, we use $n=3$ periods $\{1, 2, 4\}$.

For RISE, the default configuration uses $k=25$, patch length $p=16$, stride $s=8$, embedding dimension $d=128$, a 2-layer Transformer encoder with 4 attention heads, $G=3$ granularities with pooling widths $\{4, 2, 1\}$, and softmax temperature $\tau=0.1$.

4.2 Main Results

Table 2 summarizes the MSE and MAE performance on the eight real-world datasets across six forecast horizons, yielding 48 distinct configurations. RISE secures the highest number of first-place rankings in both metrics (23/48 for MSE and 23/48 for MAE), substantially exceeding the next strongest neural baseline (PatchTST, 8/48 in MSE). Pairwise, RISE outperforms RAFT in 71% of settings (34/48), PatchTST in 79% (38/48), iTransformer in 88% (42/48), TimeMixer in 96% (46/48), and DLinear in 100% (48/48).

The gains are most pronounced on broad equity-index datasets (S&P 500, NASDAQ, Nikkei 225, and DAX) at short-to-medium horizons ($H \leq 24$), where detrended local dynamics retain reusable temporal structure that retrieval can exploit. In contrast, RW+drift strictly dominates EUR/USD across all horizons, consistent with the Meese–Rogoff puzzle [11]. On Bitcoin, RAFT outperforms RISE

by 10.1% in average MSE, suggesting that explicit retrieval can become less reliable when extreme non-stationary volatility breaks the coherence of historical neighbors.

4.3 TFB Benchmark Evaluation

To assess external generalization, we evaluate all models on two financial-market datasets from the TFB benchmark [13] (NASDAQ and NYSE), following the official multivariate-to-multivariate (‘M’) setting and chronological 7:1:2 splits (Table 1). Forecast horizons are set to $H \in \{24, 36, 48, 60\}$, while L and top- m are selected from a broader grid for RISE and RAFT. RISE achieves the lowest overall average MSE (0.657) across the two TFB market datasets and ranks first in 7/8 MSE configurations (Table 3). On NYSE, RISE obtains the lowest MSE across all four evaluated horizons; on NASDAQ, it performs best at 3/4 horizons and remains competitive at $H=60$. These results indicate that the gains of decomposition-guided retrieval are not limited to our curated real-world suite, but also transfer to standardized public financial-market benchmarks.

4.4 Ablation Studies

We conduct two targeted ablation studies: (i) a component ablation isolating the contribution of each predictive pathway, and (ii) a diagnostic comparison of retrieval similarity keys to assess whether retrieval should be performed in the raw price space or the detrended local-dynamics space.

(i) Component ablation. We evaluate four ablated variants of RISE on six representative real-world datasets: *w/o Patch*, *w/o Trend*, *w/o Retrieval*, and *w/o Trend & Retrieval* (Table 4). All variants are retrained under the standard hyperparameter configuration.

Excluding the patch predictor incurs the most severe performance degradation (+4.7% overall increase in MSE), indicating that the patch pathway contributes most to overall predictive performance. Ablating the linear trend predictor or the retrieval pathway individually incurs a moderate penalty of 0.6–1.0%. Removing both simultaneously yields a larger combined penalty (+1.8%) than removing either pathway alone, suggesting that the two pathways provide partially non-overlapping predictive signals within the full model.

(ii) Effect of retrieval key: detrended vs. raw space. To isolate the effect of the retrieval similarity key, we conduct a diagnostic retrieval-only comparison between two variants: (a) *Raw retrieval*, which computes similarity over the raw input window, and (b) *Detrended retrieval*, which computes similarity over the causal-MA detrended signal. Both variants use the same nearest-neighbor selection and softmax-weighted aggregation rule. This analysis does not aim to reproduce the full RISE architecture; rather, it evaluates whether historical neighbors are more informative when matched in the raw price space or in the detrended local-dynamics space (Table 5).

The two variants achieve comparable prediction MSE overall, with raw retrieval being marginally lower (.130 vs. .131). However, detrended retrieval substantially improves directional accuracy (.560 vs. .537) and outperforms raw retrieval on 7 out of 8 datasets. This suggests that detrending the retrieval key better aligns historical neighbors with the future direction of the query, even when standalone point-wise MSE does not always improve. The result

Table 2: Financial forecasting results (MSE/MAE, lower is better) on eight real-world datasets with $H \in \{8, 16, 24, 36, 48, 60\}$ (48 settings). Deep-learning models are reported as multi-seed means ($n=5$) with $L=96$; for retrieval-augmented models, top- $m=20$ is used. Classical baselines are evaluated in the same z-score-normalized target space. Best in **bold red, second-best in underlined blue.**

Dataset	H	RISE		GTR		DUET		RAFT		iTrans.		PatchTST		DLinear		TimeMixer		Persist.		RW+drift		ARIMA	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
S&P 500	8	.0136	.0876	.0188	.1070	.0200	.1159	.0186	.1018	.0176	.1013	.0161	.0988	.0379	.1560	.0230	.1220	.0148	.0913	.0145	.0902	.0148	.0912
	16	.0242	.1196	.0275	.1325	.0318	.1446	.0261	.1247	.0286	.1312	.0267	.1282	.0428	.1681	.0296	.1399	.0272	.1270	.0262	.1233	.0272	.1266
	24	.0310	.1407	.0334	.1510	.0375	.1592	.0319	.1425	.0343	.1489	.0327	.1457	.0484	.1774	.0352	.1552	.0354	.1500	.0329	.1431	.0354	.1493
	36	.0426	.1661	.0441	.1743	.0472	.1780	.0431	.1673	.0474	.1749	.0425	.1660	.0554	.1892	.0479	.1811	.0507	.1809	.0444	.1697	.0504	.1799
	48	.0550	.1893	.0576	.1970	.0584	.1982	.0558	.1903	.0578	.1931	.0524	.1840	.0616	.1985	.0624	.2034	.0692	.2103	.0575	.1927	.0687	.2090
	60	.0706	.2152	.0723	.2183	.0699	.2189	.0705	.2147	.0667	.2081	.0619	.2013	.0723	.2161	.0758	.2253	.0924	.2424	.0734	.2158	.0916	.2409
	Avg	.0395	.1531	.0423	.1633	.0441	.1691	.0410	.1569	.0421	.1596	.0387	.1540	.0531	.1842	.0457	.1712	.0483	.1670	.0415	.1558	.0480	.1662
NASDAQ	8	.0137	.0882	.0190	.1055	.0208	.1154	.0189	.1024	.0206	.1080	.0160	.0961	.0424	.1655	.0227	.1175	.0153	.0935	.0152	.0927	.0155	.0941
	16	.0246	.1184	.0277	.1279	.0336	.1452	.0267	.1239	.0325	.1372	.0273	.1243	.0482	.1788	.0306	.1352	.0283	.1273	.0276	.1244	.0286	.1279
	24	.0306	.1357	.0334	.1441	.0385	.1592	.0315	.1377	.0388	.1549	.0314	.1373	.0532	.1875	.0357	.1502	.0351	.1453	.0333	.1396	.0352	.1452
	36	.0408	.1599	.0452	.1710	.0482	.1787	.0416	.1615	.0483	.1765	.0376	.1531	.0598	.1989	.0479	.1768	.0486	.1739	.0438	.1646	.0486	.1735
	48	.0520	.1823	.0588	.1956	.0553	.1925	.0530	.1840	.0554	.1902	.0436	.1665	.0645	.2058	.0589	.1959	.0652	.2026	.0563	.1883	.0651	.2020
	60	.0651	.2052	.0713	.2160	.0609	.2034	.0646	.2045	.0607	.2010	.0483	.1763	.0743	.2217	.0672	.2119	.0839	.2308	.0690	.2095	.0838	.2302
	Avg	.0378	.1483	.0426	.1600	.0429	.1657	.0394	.1523	.0427	.1613	.0340	.1423	.0570	.1930	.0438	.1646	.0461	.1622	.0409	.1532	.0461	.1621
Nikkei 225	8	.0668	.1836	.0903	.2269	.0831	.2160	.0773	.2016	.0837	.2192	.0764	.2077	.1709	.3106	.1110	.2571	.0748	.1924	.0747	.1920	.0880	.1947
	16	.1160	.2446	.1362	.2794	.1314	.2756	.1213	.2521	.1487	.2970	.1363	.2814	.2250	.3546	.1515	.2942	.1295	.2541	.1288	.2532	.1467	.2533
	24	.1558	.2842	.1763	.3227	.1781	.3228	.1575	.2863	.1977	.3462	.1968	.3443	.2791	.3952	.1808	.3206	.1681	.2906	.1662	.2884	.1906	.2916
	36	.2116	.3350	.2424	.3810	.2485	.3852	.2118	.3354	.2574	.4002	.2734	.4124	.3581	.4472	.2393	.3712	.2166	.3380	.2118	.3337	.2486	.3422
	48	.2724	.3771	.3046	.4204	.3184	.4304	.2789	.3829	.3191	.4438	.3425	.4576	.4341	.4915	.3115	.4170	.2764	.3810	.2674	.3753	.3242	.3937
	60	.3281	.4118	.3589	.4491	.3875	.4695	.3371	.4204	.3788	.4799	.3964	.4874	.5274	.5460	.3771	.4566	.3361	.4183	.3209	.4101	.3982	.4369
	Avg	.1918	.3060	.2181	.3466	.2245	.3499	.1973	.3131	.2309	.3644	.2370	.3652	.3324	.4242	.2285	.3528	.2003	.3124	<u>.1950</u>	<u>.3088</u>	.2327	.3187
Gold	8	.0285	.1260	.0397	.1520	.0478	.1672	.0339	.1425	.0376	.1497	.0322	.1355	.1147	.2642	.0512	.1751	.0311	.1331	.0304	.1317	.0323	.1353
	16	.0524	.1695	.0622	.1884	.0758	.2089	.0554	.1764	.0662	.1973	.0560	.1776	.1507	.2993	.0742	.2100	.0597	.1819	.0572	.1788	.0636	.1882
	24	.0731	.2006	.0858	.2214	.1063	.2461	.0747	.2035	.0944	.2361	.0798	.2131	.1902	.3358	.0977	.2409	.0859	.2192	.0803	.2129	.0912	.2277
	36	.1083	.2427	.1265	.2665	.1507	.2894	.1112	.2457	.1424	.2864	.1190	.2578	.2477	.3842	.1368	.2811	.1361	.2736	.1236	.2625	.1442	.2842
	48	.1439	.2790	.1586	.2975	.1944	.3287	.1500	.2836	.1800	.3194	.1581	.2938	.2874	.4145	.1686	.3098	.1880	.3201	.1668	.3028	.2001	.3334
	60	.1721	.3057	.1854	.3226	.2284	.3601	.1808	.3130	.1988	.3378	.1787	.3154	.3606	.4689	.1881	.3292	.2310	.3563	.1990	.3308	.2457	.3716
	Avg	.0964	.2206	.1097	.2414	.1339	.2667	<u>.1010</u>	<u>.2274</u>	.1199	.2545	.1040	.2323	.2252	.3611	.1194	.2577	.1220	.2473	.1096	.2366	.1295	.2567
EUR/USD	8	.0462	.1608	.0681	.2053	.0600	.1857	.0594	.1870	.0551	.1762	.0516	.1732	.1375	.2995	.0804	.2237	.0415	.1527	.0415	.1524	.0423	.1548
	16	.0823	.2217	.1071	.2615	.1006	.2496	.0840	.2260	.0984	.2413	.0929	.2387	.1704	.3314	.1174	.2748	.0719	.2073	.0716	.2067	.0732	.2091
	24	.1216	.2724	.1468	.3078	.1452	.3076	.1139	.2656	.1436	.2971	.1353	.2931	.2154	.3702	.1498	.3127	.1019	.2487	.1011	.2477	.1033	.2499
	36	.1725	.3253	.2020	.3590	.1978	.3600	.1611	.3166	.2093	.3610	.1992	.3547	.2823	.4214	.1982	.3574	.1490	.3034	.1469	.3017	.1499	.3035
	48	.2280	.3711	.2470	.3940	.2423	.3936	.2096	.3561	.2630	.4009	.2617	.4026	.3620	.4724	.2486	.3938	.2054	.3486	.2016	.3460	.2058	.3486
	60	.2682	.4014	.2843	.4177	.2751	.4113	.2508	.3869	.3034	.4281	.3205	.4425	.4380	.5143	.2887	.4199	.2554	.3863	.2496	.3824	.2550	.3856
	Avg	.1531	.2921	.1759	.3242	.1702	.3179	.1465	.2897	.1788	.3174	.1769	.3175	.2676	.4015	.1805	.3304	<u>.1375</u>	<u>.2745</u>	.1354	.2728	.1382	.2753
Bitcoin	8	.0390	.1318	.0530	.1593	.0751	.1856	.0436	.1460	.0681	.1849	.0483	.1513	.1430	.2636	.0681	.1860	.0437	.1407	.0434	.1404	.0459	.1439
	16	.0704	.1785	.0911	.2099	.1271	.2467	.0734	.1832	.1320	.2580	.0895	.2069	.1966	.3031	.1095	.2351	.0809	.1920	.0800	.1912	.0862	.1989
	24	.1067	.2216	.1300	.2521	.1756	.2902	.1090	.2236	.1829	.3062	.1382	.2571	.2470	.3363	.1448	.2683	.1235	.2391	.1216	.2377	.1294	.2453
	36	.1669	.2769	.1970	.3068	.2580	.3465	.1647	.2752	.2638	.3624	.2278	.3220	.3297	.3839	.2120	.3217	.1891	.2928	.1845	.2898	.1982	.3002
	48	.2515	.3366	.2740	.3614	.3329	.3929	.2282	.3222	.3652	.4261	.3313	.3821	.4132	.4286	.2815	.3697	.2663	.3428	.2576	.3379	.2825	.3535
	60	.3709	.4088	.3500	.4074	.4127	.4407	.2847	.3575	.4324	.4627	.4479	.4471	.4992	.4729	.3655	.4189	.3376	.3847	.3232	.3779	.3567	.3967
	Avg	<u>.1676</u>	<u>.2590</u>	.1825	.2828	.2302	.3171	.1506	.2513	.2407	.3334	.2138	.2944	.3048	.3647	.1969	.3000	.1735	.2654	.1684	.2625	.1831	.2731
Hang Seng	8	.0492	.1588	.0719	.1893	.0613	.1741	.0632	.1864	.0670	.1888	.0569	.1705	.1746	.3528	.0918	.2157	<u>.0509</u>	.1581	.0509	.1581	.0628	.1638
	16	.0926	.2090	.1155	.2364	.1002	.2196	.0973	.2172	.1182	.2423	.1027	.2193	.2160	.3982	.1329	.2567	.1006	.2131	.1006	.2131	.1333	.2264
	24	.1309	.2488	.1553	.2765	.1359	.2623	.1303	.2497	.1569	.2807	.1441	.2620	.2630	.4440	.1652	.2840	.1458	.2591	.1458	.2590	.1993	.2772
	36	.1749	.2924	.1968	.3175	.1799	.3076	.1684	.2868	.2077	.3248	.1967	.3095	.3464	.5154	.2067	.3194	.1978	.3059	.1979	.3058	.2909	.3304
	48	.2164	.3298	.2245	.3466	.2036	.3365	.1864	.3109	.2335	.3484	.2244	.3390	.4448	.5800	.2203	.3381	.2259	.3385	.2261	.3385	.3555	.3686
	60	.2750	.3746	.2538	.3754	.2410	.3731	.2190	.3445	.2781	.3804	.2711	.3746	.5626	.6459	.2539	.3729	.2580	.3736	.2582	.3736	.4199	.4084
	Avg	.1565																					

Table 3: TFB benchmark results (MSE/MAE, lower is better) evaluated on two financial-market datasets for $H \in \{24, 36, 48, 60\}$. Deep-learning results report multi-seed averages ($n=5$); classical baselines operate deterministically per window. Best performance in **bold red, second-best in underlined blue.**

Dataset	H	RISE		GTR		DUET		RAFT		iTrans.		PatchTST		DLinear		TimeMixer		Persist.		RW+drift		ARIMA	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
NASDAQ	24	.553	<u>.531</u>	.738	.658	.566	.566	.570	.544	<u>.558</u>	.560	.621	.570	1.634	.971	1.194	.819	.684	.582	.666	.575	.662	.531
	36	.809	.655	1.084	.783	.866	.686	.897	.685	<u>.844</u>	<u>.685</u>	.989	.714	2.012	1.061	1.250	.849	1.180	.776	1.138	.762	1.140	.719
	48	1.077	.750	1.299	.843	1.177	.774	1.206	.788	<u>1.120</u>	<u>.770</u>	1.299	.800	2.101	1.080	1.343	.865	1.720	.938	1.653	.918	1.840	.928
	60	1.259	<u>.811</u>	1.317	.845	<u>1.254</u>	.803	1.374	.841	1.349	.840	1.376	.827	1.949	1.051	1.139	.820	2.082	1.029	1.999	1.006	2.537	1.090
	Avg	.925	.687	1.110	.782	<u>.966</u>	<u>.707</u>	1.012	.715	.968	.714	1.071	.728	1.924	1.041	1.231	.839	1.417	.831	1.364	.815	1.545	.817
NYSE	24	.183	.267	.266	.342	.240	.312	.213	.298	.247	.325	.207	.287	.582	.571	.502	.522	.213	.276	.193	.260	<u>.187</u>	<u>.263</u>
	36	.309	.342	.408	.424	.410	.419	.353	.382	.397	.415	.345	.375	.854	.707	.773	.645	.377	.374	<u>.330</u>	<u>.343</u>	.360	.371
	48	.446	<u>.421</u>	.559	.492	.610	.525	.494	.448	.546	.481	.492	.447	1.191	.846	1.049	.785	.572	.468	<u>.485</u>	.420	.564	.471
	60	.617	.511	.750	.581	.849	.643	.668	.537	.724	.566	<u>.664</u>	.537	1.576	.984	1.250	.887	.824	.591	.677	<u>.522</u>	.824	.594
	Avg	.389	.386	.496	.460	.527	.475	.432	.416	.479	.447	.427	.412	1.051	.777	.894	.710	.496	.428	<u>.421</u>	<u>.386</u>	.484	.425
Overall Avg		.657	.537	.803	.621	.747	.591	<u>.722</u>	<u>.566</u>	.724	.581	.749	.570	1.488	.909	1.063	.775	.957	.630	.893	.601	1.015	.621
1st Count		7	4	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	2	0	1

Table 4: Component ablation (avg. MSE across $H \in \{8, 16, 24, 36, 48, 60\}$). Best in **bold red, second-best in underlined blue. Δ : relative change vs. Full RISE.**

Dataset	Full RISE	w/o Patch	w/o Trend	w/o Retrieval	w/o Trend & Retrieval
S&P 500	.0380	.0435	.0382	.0381	.0380
NASDAQ	.0367	.0417	.0376	<u>.0374</u>	.0383
Nikkei 225	.1879	.1957	<u>.1882</u>	.1884	.1902
Gold	.0950	.1025	<u>.0963</u>	.0966	.0988
EUR/USD	.1469	.1473	<u>.1473</u>	.1508	.1500
Bitcoin	<u>.1450</u>	.1491	.1457	.1447	.1461
Overall	.1083	.1133	<u>.1089</u>	.1093	.1102
Δ vs. Full	—	+4.7%	+0.6%	+1.0%	+1.8%

Table 5: Retrieval-key comparison using raw windows vs. causal-MA detrended windows as the similarity space. Both variants use the same nearest-neighbor selection and softmax-weighted aggregation rule. Directional accuracy ($\uparrow$): fraction of forecast steps whose predicted direction from the last observed value matches the ground truth. Prediction MSE ($\downarrow$): lower is better. Results averaged over $H \in \{8, \dots, 60\}$. Bold: better of the two per metric.

Dataset	Prediction MSE ($\downarrow$)		Directional accuracy ($\uparrow$)	
	Raw	Detrended	Raw	Detrended
S&P 500	.055	.039	.516	.591
NASDAQ	.039	.038	.601	.606
Nikkei 225	.182	.204	.558	.525
Hang Seng	.202	.217	.540	.544
DAX	.111	.119	.537	.580
Gold	.102	.095	.531	.549
EUR/USD	.204	.185	.497	.557
Bitcoin	.147	.150	.516	.531
Overall	.130	.131	.537	.560

Table 6: Hyperparameter sensitivity. Kernel k : S&P 500, averaged over $H \in \{24, 36, 48, 60\}$. Look-back L and top- m : average MSE across six datasets $\times$ four horizons ($H \in \{24, 36, 48, 60\}$). All runs use the OHLCV-only main configuration; bold marks the default ($L=96$, top- $m=20$, $k=25$) used in the main results.

Kernel k	MSE	L	Avg MSE	top- m	Avg MSE
5	.0495	36	.1409	1	.1477
15	.0496	64	.1417	5	.1464
25	.0494	96	.1466	10	.1462
45	.0491	104	.1512	20	.1463
65	.0492				
Range: 1.0%		Range: 7.3%		Range: 1.0%	

Kernel size (k): Performance varies within a 1.0% range, indicating low sensitivity to kernel size. We adopt $k=25$ as the default, corresponding approximately to one active trading month of daily observations. **Look-back window (L):** Shorter context windows ($L=36$) outperform extended look-backs, with a range of 7.3% across tested values. This suggests that excessively distant historical observations may precede structural regime changes, introducing noise that degrades predictive accuracy. **Top- m selection:** Predictive accuracy remains stable across different values of m (range: 1.0%), indicating that RISE is not sensitive to the number of retrieved neighbors.

5 Conclusion

We introduced RISE, a decomposition-guided retrieval architecture for financial time-series forecasting. RISE uses a strictly causal trend-detrended decomposition as a routing mechanism: the macro trend is directed to a channel-independent linear head, while the detrended residual is processed in parallel by a multi-granularity similarity-based retrieval module and a patch-based Transformer, with the three pathway outputs combined via a learnable weighted fusion. The retrieved detrended futures are also injected as Retrieval-as-Context (RAC) tokens into the patch encoder, allowing retrieval to shape representation learning rather than acting only through

output-level fusion. RAC adds only $G(H + 2)d$ parameters (3.8K–23.8K for $G=3$, $d=128$).

Empirical evidence. Across eight real-world financial datasets, RISE attains the highest first-place count in both MSE and MAE (23/48 each), nearly triple the next best neural baseline, and pairwise outperforms all deep-learning baselines on 71–100% of settings. On the independent TFB financial-market benchmark (NASDAQ, NYSE), RISE attains the lowest overall MSE (.657) and ranks first in 7/8 MSE configurations. A retrieval-key diagnostic (Table 5) further isolates the contribution of decomposition: while raw and detrended retrieval are essentially tied on aggregate MSE (.130 vs. .131), detrended retrieval improves directional accuracy on seven of eight datasets (.560 vs. .537), directly substantiating the motivating example (Figure 1). Component ablations confirm that the patch, trend, and retrieval pathways provide partially non-overlapping signals; RISE is robust to retrieval hyperparameters ($\leq 1.0\%$ variation), while a shorter look-back ($L=36$) further reduces MSE by 3.9% over our default $L=96$, indicating that the reported margins under protocol-parity are conservative.

Limitations. RISE’s predictive advantage diminishes in regimes where historical similarity offers limited actionable signal: a naive drift baseline dominates EUR/USD and long-horizon Hang Seng (consistent with the Meese–Rogoff puzzle), and RAFT’s raw-signal retrieval outperforms RISE on Bitcoin’s volatility-burst regime by 10.1% in average MSE at long horizons, yielding a slightly lower Overall Avg MSE for RAFT (.1198 vs. .1221). Our dataset suite also under-represents mainland-Chinese equities, energy commodities, and broader FX pairs. These limitations motivate learnable asset-adaptive decomposition filters and robust retrieval protocols for near-random-walk and high-volatility regimes, as well as extending RISE to intraday and microstructural forecasting.

More broadly, RISE suggests that decomposition in retrieval-augmented forecasting is best understood as a division of labor: retrieval over the detrended residual specializes in directional structure, while a separate trend pathway handles absolute level. The empirical effectiveness of this design is not visible in any single retrieval metric alone, but emerges once directional accuracy and integration with a learned representation are considered jointly.

GenAI Usage Disclosure

During the preparation of this work, the authors utilized a generative AI tool (Claude Sonnet 4.6) to polish the manuscript and source code. Specifically, the tool was used to refine the implementation of the GPU processing part in the code, as well as to correct wording and enhance sentence clarity throughout the entire paper. No part of the research design, experimental execution, data analysis, or interpretation of results was conducted using generative AI. All core technical and scientific contributions remain entirely the authors’ own work.

Acknowledgments

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